Kinetic surrogates for techno-economic screening of ethane cracking

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Abstract

Ethane-cracker operating conditions determine both the single-pass ethylene yield and the gas flow processed per tonne of ethylene. We quantify how the gas temperature–time profile in the cracking coil changes process loads, production cost, and carbon emissions by combining detailed kinetics with an open-source process model. At a coil outlet temperature (COT) of 850 °C and 0.35 s residence time, delaying heating over the tested range of profiles increases the production cost with fired heating from 456 to 806 USD t^{-1} at fixed process and price assumptions. In the COT–residence-time scan, a residence time longer than that of maximum yield decreases both the single-pass ethylene yield and the production cost because lower heating cost and higher coproduct credits outweigh additional feed and processing costs. Direct Cantera calculations reproduce this ordering, and comparisons with four published process cases quantify how differences in effluent composition change downstream costs. At the same coil effluent composition, coil duty, and equipment costs, electric heating gives a lower production cost than fired heating below a break-even electricity price of 2.28 cents/kWh, for natural gas at 4 USD/GJ and heat-delivery efficiencies of 95% and 60%, respectively.

1 Introduction

Ethane cracking couples high-temperature reaction chemistry to energy-intensive product recovery. Changes in conversion and selectivity alter the amounts of unreacted ethane and coproducts that must be compressed, chilled, and separated per tonne of ethylene. Chen et al. quantify this downstream load: their oxidative-dehydrogenation process has a lower minimum selling price than steam cracking but higher life-cycle emissions, with refrigeration contributing 1.30 kg CO_2e per kilogram of ethylene[1]. Shin et al. show that the economics and emissions of cracker electrification depend on electricity supply and the treatment of hydrogen-rich coproducts[2]. Shortcut column models and separation-network synthesis provide methods for evaluating composition-dependent recovery loads during process design[4,5].

Internal electric heating provides additional control over heat delivery. Balakotaiah and Ratnakar analyze resistance-heated reactor configurations in terms of electrical supply, transport, and reaction time scales[7]. Implementations include catalyst-coated SiSiC foams for methane reforming, FeCrAl coils for catalytic ethane dehydrogenation, and induction-heated ceramic foams for reverse water-gas shift[8–10]. These configurations generate heat within the reactor, with different solid-to-gas heat-transfer and loss mechanisms. Carbon-fiber-paper reactor calculations resolve localized heating and buoyancy-driven flow[11], while material screening quantifies how electrical and thermal properties constrain heating rates and temperature uniformity[12]. Mittal et al. combined kinetic cal-

culations, transport modeling, and carbon-fiber-paper experiments to examine short-contact-time ethane cracking. Their process analysis calculated energy demand and process economics from the predicted product distribution, and the economic result depended on electricity supply[13].

Cattry et al. also combine internally resistance-heated and fired reactor models with process simulation, economics, and life-cycle assessment[3]. Their reactor design with the highest ethylene yield differs from the minimum-cost design because added heating intensity increases electricity demand. A recent steam-cracker retrofit preprint similarly combines kinetics, process simulation, and optimization to evaluate olefin yield and coproduct value[6]. These studies establish the need to assess reactor improvements against the cost of the full process. Here, we examine how the time at high temperature changes fresh-feed demand, ethane recycle, separation loads, and production cost. The comparisons use the same downstream correlations and price assumptions, so differences in these quantities result from the coil effluent composition and coil duty.

Surrogate models make repeated evaluation of detailed kinetics practical. Bradley et al. describe ways to combine first-principles and data-driven submodels[14], and Döppel and Votsmeier demonstrate surrogate acceleration of surface kinetics[15]. Gaussian-process (GP) models also support computational searches, as implemented in NEXTorch[16]. We use a GP representation of AramcoMech 3.0 calculations in Cantera to examine coil outlet temperature (COT), residence time, dilution steam ratio (kg steam per kg ethane), pressure, and the gas temperature–time profile[17,18]. Changing the gas temperature–time profile changes the coil effluent composition and therefore the downstream loads, whereas changing the heat source at the same operating condition isolates the effect of the heat source. The calculations show how delayed heating increases the flow processed per tonne of ethylene and how heating cost and coproduct credits place the minimum-cost condition at a longer residence time than the maximum-yield condition. Calibration of process parameters to published utility demands quantifies how assumed utility discrepancies change the break-even electricity price. The released code, training data, and process equations allow users to inspect each assumption and replace individual submodels (**Figure 1**).

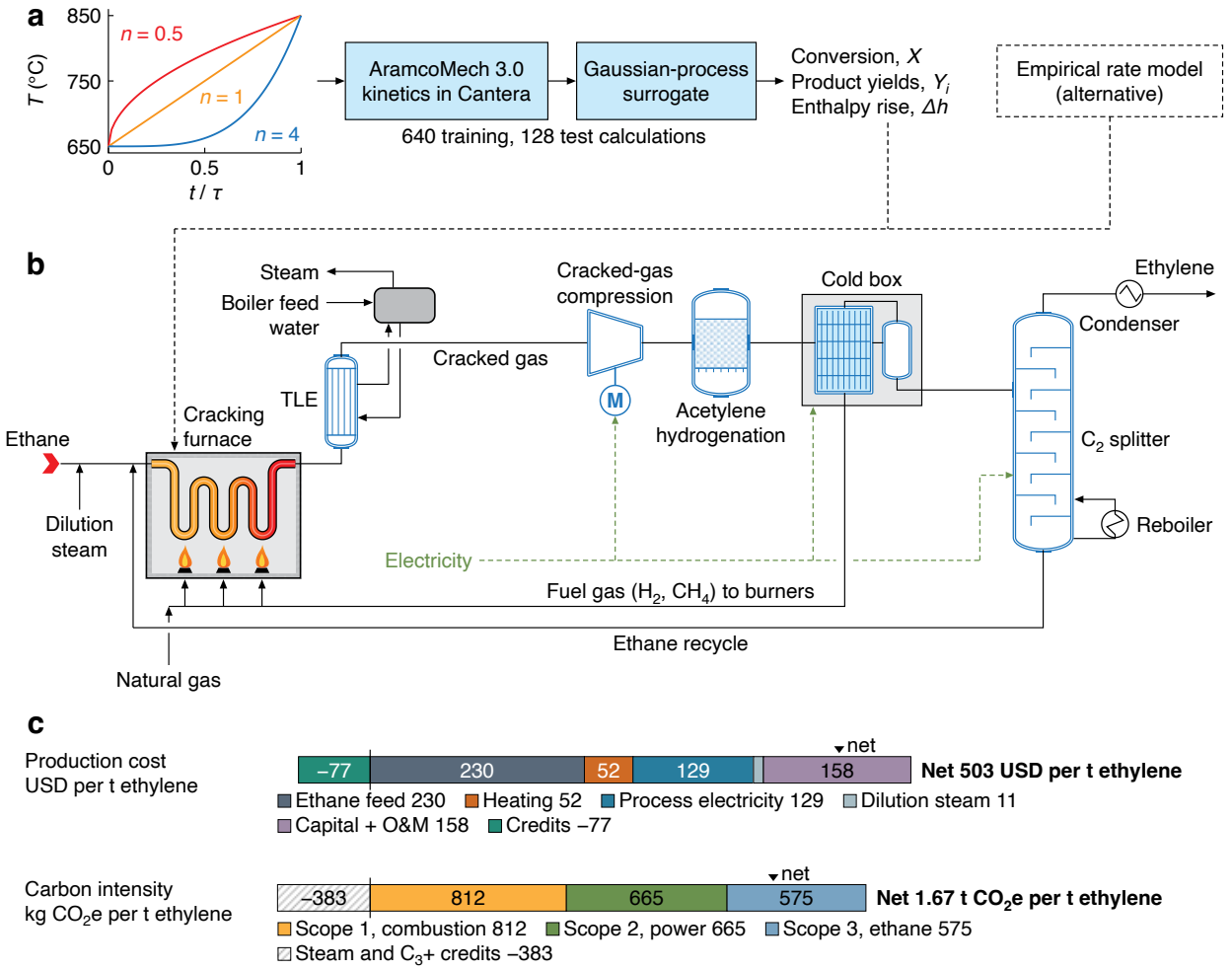


Figure 1. Reaction-to-process screening workflow. **(a)** Prescribed gas temperature–time profiles at 850 °C COT, with heating exponents of 0.5, 1, and 4. A GP surrogate trained on 640 AramcoMech–Cantera calculations and tested on 128 independent calculations predicts single-pass conversion, product yields, and enthalpy rise. An empirical rate model calculates the same outputs for the process model. **(b)** Process schematic of the fired case showing cracking, transfer-line-exchanger (TLE) heat recovery to steam, cracked-gas compression, acetylene hydrogenation, cold separation in the cold box (chilling train and demethanizer), C_2 splitting, and ethane recycle. The furnace burns fuel gas (methane and hydrogen recovered in the cold box) before purchased natural gas. Compression, cold-box refrigeration, and the C_2 splitter use electricity. Solid arrows denote material or energy flows, black dashed arrows denote calculation inputs, and green dashed lines denote electricity supply. **(c)** Production cost and carbon intensity by component, per tonne of ethylene. Hatched segments are steam and coproduct credits, and markers show the net values. Values are calculated for the reference fired case at 850 °C, 0.35 s, 1.5 bar, dilution steam ratio 0.35 kg steam per kg ethane, and heating exponent 1.

2 Methods

2.1 Comparison basis and calculation sequence

The model screens operating conditions and heating options at a common nominal ethylene capacity. Material flows, utility demands, and costs are reported per unit mass of final ethylene after acetylene hydrogenation. Each case specifies COT, residence time, dilution steam ratio, reactor pressure, and gas temperature–time profile. The kinetic calculation gives single-pass conversion, product yields, and enthalpy rise. The process model then calculates fresh feed, recycle, utility loads, and section costs. Prices, equipment correlations, and product specifications remain fixed unless explicitly varied. Table 1 lists the reference inputs. Run length, the operating time between shutdowns, determines annual production and the fixed charges per tonne.

2.2 Reactor calculations and operating domain

The gas temperature–time profile is specified as an input so that early and delayed heating can be compared at the same COT and residence time. The reactor calculation integrates the species balances of a constant-pressure fluid element over this profile:

$$\begin{aligned} T(f) &= T_{\text{in}} + (T_{\text{out}} - T_{\text{in}}) f^n \\ f &= t/\tau \end{aligned} \tag{1}$$

Here t is elapsed time and τ is residence time. The inlet temperature T_{in} is 650 °C, and T_{out} is the COT. The heating exponent n sets the profile shape: $n = 1$ corresponds to a linear temperature rise, $n < 1$ heats the gas earlier and holds it longer near the COT, and $n > 1$ delays heating. Product composition is calculated over this profile, and the enthalpy change is evaluated from the inlet and outlet compositions and temperatures. Prescribed-temperature studies of periodically operated reaction networks show how exposure time at high temperature controls desired and secondary chemistry[19]. Here, $T(t)$ is the temperature of one fluid element as a function of its time in the reactor.

The training domain covers COTs of 750–1000 °C, residence times of 0.02–1.5 s, dilution steam ratios d of 0–0.70 kg steam per kg ethane, pressures of 1–5 bar, and heating exponents of 0.45–4. Residence time is sampled logarithmically. Each trajectory uses 20 constant-temperature segments, with boundaries uniformly spaced in f^n. Cantera 3.2.0 integrates the chemistry with relative and absolute tolerances of 10^{-8} and 10^{-15}, respectively. The converted AramcoMech 3.0 mechanism contains 581 species and 3037 reactions.

2.3 Surrogate construction and validation

The training set contains 512 operating conditions across the full domain and 128 additional high-severity conditions, sampled using Latin hypercubes. The high-severity conditions cover 900–1000 °C, 0.2–1.5 s, and dilution steam ratios of 0–0.40, with the same pressure and heating-exponent ranges. An independent test set of 128 conditions across the full domain is excluded from training. GP regression[20] uses the inputs $[T_{\text{out}}, \log_{10}\tau, d, P, n]$, where P is pressure. Inputs are normalized using the observed training extrema. Conversion and molar ethylene selectivity use logit transforms, and species yields and enthalpy rise use logarithmic transforms. An anisotropic radial-basis-function kernel is shared across outputs. Kernel selection and uncertainty calibration use an internal fitting/tuning split. The selected kernel is then fitted to all 640 training conditions. SI Section S1.3 gives the fitting procedure.

The surrogate predicts single-pass ethane conversion, ethylene selectivity, seven single-pass mass yields (ethylene, methane, hydrogen, acetylene, ethane, C_3, and C_{4+}), and enthalpy rise. The C_3 and C_{4+} groups sum species with three and at least four carbon atoms, respectively. The surrogate is evaluated against three criteria: $R^2 \geq 0.90$, root-mean-square error divided by the observed test-set range below 0.08, and at least 80% empirical coverage by the nominal 95% prediction intervals for each output. These checks measure how closely the surrogate reproduces the kinetic calculations.

Raw Cantera calculations are checked for carbon, hydrogen, oxygen, and mass closure. A separate five-condition comparison of 20 and 40 segments tests temporal discretization. The GP predictions are compared with the independent test set, and the material and energy balances are checked against the raw Cantera outputs.

2.4 Conversion to a product basis

The coil processes fresh and recycled ethane together. Lower single-pass ethylene yield therefore increases the coil feed required per unit product, and lower single-pass conversion increases the unreacted ethane recycled to the coil. Let X be the single-pass ethane conversion and Y_E the single-pass ethylene yield (kg ethylene per kg coil-inlet ethane). The acetylene hydrogenation reactor adds ethylene and ethane yields ΔY_E and ΔY_A on the same basis. With complete ethane recovery, the flows per unit mass of final ethylene are

$$\begin{aligned} Y_{E,\text{final}} &= Y_E + \Delta Y_E \\ m_{\text{coil}} &= \frac{1}{Y_{E,\text{final}}} \\ m_{\text{fresh}} &= \max(0, X - \Delta Y_A) m_{\text{coil}} \\ m_{\text{recycle}} &= m_{\text{coil}} - m_{\text{fresh}} \end{aligned} \tag{2}$$

Species yields after acetylene hydrogenation are divided by $Y_{E,\text{final}}$ to express them per unit mass of final ethylene. The ultimate ethylene yield, Y_E/X (kg ethylene per kg ethane consumed), and the molar ethylene selectivity are derived from Y_E and X, and the independently predicted selectivity is used as a consistency check. The empirical rate model predicts no acetylene and uses $\Delta Y_E = \Delta Y_A = 0$.

The total reactor duty includes feed preheating and cracking: a constant-heat-capacity calculation covers 150–650 °C, and Cantera calculates the enthalpy rise from 650 °C to the COT. An assumed transfer-line-exchanger (TLE) recovery of 70% of the calculated sensible heat is valued at the specified export-steam price. Dilution-steam generation and quench condensation add a separate charge. These assumptions account for heat purchase and recovery without solving a plant heat-exchanger network.

2.5 Process and economic calculation

Water is removed before four-stage compression of the dry cracked gas from a fixed suction pressure of 1.5 bar to 32 bar. After compression, the acetylene hydrogenation reactor reduces acetylene to 5 mol ppm in the C_2 fraction, with 90% selectivity to ethylene and 10% to ethane. Its hydrogen consumption reduces the fuel gas available for heating or export. The cold box removes the fuel gas, and the C_2 splitter separates ethylene from the ethane recycled to the coil. Refrigeration work depends on gas flow, the fuel-gas (methane plus hydrogen) mole fraction, and the assumed refrigeration efficiency. The splitter load depends on the ethylene/ethane ratio and

specified ethylene purity. The cold-box recovery target sets its cooling requirement. Product flows use the complete-recovery basis in Eq. (2). The equations and constants are summarized in the Supporting Information (SI).

Section-specific scaling of coil flow, compressor work, refrigeration work, and splitter load sets the bare-module cost relative to a reference bare-module cost of 126.8 million USD at 610 kt y^{-1}. The total capital investment (TCI) is 2.5 times the scaled bare-module cost. Annual capital and operating charges are divided by annual production corrected for the on-stream factor from the run-length correlation. The production cost C_E (USD per t ethylene) is the levelized cost of ethylene, including the annualized capital charge and coproduct credits:

$$C_E = C_{\text{feed}} + C_{\text{heat}} + C_{\text{process electricity}} + C_{\text{dilution}} + C_{\text{capital}} + C_{\text{O\&M}} - C_{\text{credits}} \tag{3}$$

The credits are for TLE steam, exported fuel gas, and C_3+ coproducts at the prices in Table 1. This production cost is not a minimum selling price. All monetary inputs are fixed scenario values in USD without inflation or location adjustment. The reference bare-module cost is derived from the published investment for the conventional case[13]. Section cost fractions, scaling exponents, and shortcut utility correlations are model assumptions.

Table 1. Reference screening conditions. One variable is changed at a time unless stated otherwise.

Input	Value
Coil outlet temperature (COT)	850 °C
Residence time	0.35 s
Dilution steam ratio, d (kg steam per kg ethane)	0.35
Reactor pressure	1.5 bar
Heating exponent	1
Nominal ethylene capacity	610 kt y^{-1}
Fired / electric heat-delivery efficiency	0.60 / 0.95
Ethane price	200 USD t^{-1}
Natural-gas price	4 USD GJ^{-1}
Electricity price	7.0 cents/kWh
Grid emission factor	0.36 kg CO_2e kWh^{-1}
Export-steam credit	9.8 USD GJ^{-1}
C_3+ credit	550 USD t^{-1}
Annual capital charge	0.10
Labor	5 million USD y^{-1}
Ethane upstream emission factor	0.50 kg CO_2e kg^{-1} ethane

2.6 Fired and electric heating at the same operating condition, and carbon accounting

The heating comparison holds the gas temperature–time profile, coil effluent composition, process loads, and capital scaling fixed. With fired heating, the furnace burns the available fuel gas before purchased natural gas. With electric heating, the coil heat is purchased as electricity, and the exported fuel gas receives a credit at the natural-gas price on a lower-heating-value (LHV) basis. The heat-delivery efficiency changes from 0.60 to 0.95. This comparison quantifies the cost of

changing the heat source at fixed coil effluent composition and coil duty. The separate study of the gas temperature–time profile evaluates changes in the coil effluent composition.

The carbon calculation applies emission factors to purchased natural gas, eventual combustion of the fuel gas, process and heating electricity, and upstream ethane. TLE steam and C_3+ receive displacement credits. All plotted cases count eventual combustion of the full methane coproduct, including methane not burned in the cracking furnace. The production cost credits this exported fuel gas at the natural-gas price, and the carbon intensity includes its combustion emissions. SI Section S3 specifies the emission factors and system boundary.

2.7 Literature calibration and paired cost uncertainty

We use two published utility demands to calibrate effective process parameters. For Chen et al., operating conditions are first selected to match the reported residual ethane fraction, and compressor power per unit ethylene production then calibrates a compression multiplier. For Shin et al., fresh-feed demand sets the operating conditions, and total combustion energy calibrates fired efficiency. The calibration assumes uniform parameter priors and equal prior weights for the selected operating conditions. Differences between calculated and reported utilities are represented by a Gaussian distribution with an assumed standard deviation of 5%, 10%, or 20% of the reported value. Each parameter integral uses 161 quadrature nodes.

For each assumed discrepancy level, 512 parameter sets are evaluated for both heating modes at the reference operating condition. Each pair uses the same compression, heat-recovery, and shared capital factors, GP predictions, and prices. Electric heating efficiency and its furnace-capital factor are varied in separate scenarios. The resulting cost distributions quantify uncertainty from these process parameters while holding the GP predictions fixed. GP prediction uncertainty is evaluated separately with the independent test set. SI Section S1.4 defines the priors, weighting, and paired calculations.

3 Results and discussion

3.1 Accuracy of the kinetic surrogate

For the 128 Cantera calculations excluded from training, the GP predicts all ten outputs with root-mean-square errors below 1.62% of each output's observed range. Ethane conversion, single-pass ethylene yield, and enthalpy rise have R^2 values of 0.9997, 0.9994, and 0.9993, respectively, with normalized errors of 0.59%, 0.85%, and 0.77%. The largest error occurs for the C_3 lump. **Figure S1** shows representative parity plots, and Table S1 lists the metrics for all outputs.

Across all 768 raw kinetic calculations, the largest elemental residual is 5.69×10^{-11} and the largest total-mass residual is 2.00×10^{-15}. For five test conditions, doubling the number of temperature segments keeps changes in conversion, selectivity, and yield below 0.005 and changes in enthalpy below 1%. At the reference condition, increasing from 20 to 40 segments changes conversion by 0.088 percentage points and enthalpy by 0.148%.

Process screening depends on ratios of kinetic outputs. Applying Eq. (2) to the independent test set gives a median absolute relative error of 0.665% in coil ethane flow and a maximum of 15.9%. At low single-pass ethylene yield, a small yield error can cause a much larger error in the required coil flow. At the reference condition, the process-derived and directly predicted molar selectivities differ by 0.669 percentage points. The process calculations use selectivity derived from yield and conversion.

3.2 Operating conditions, recycle, and production cost

The GP reference case predicts 38.0% single-pass ethane conversion, a single-pass ethylene yield of 0.329 kg per kg coil-inlet ethane, and 1.88 kg of recycled ethane per kg ethylene. The empirical reference case predicts 65.0% conversion and 0.656 kg recycled ethane per kg ethylene (Table 2). The empirical rate expression is calibrated at the COT, whereas the GP case uses the 650–850 °C gas temperature–time profile. The two cases therefore differ in both the kinetic model and the gas temperature–time profile.

Table 2. Calculated reference cases. The GP fired and electric cases use the same operating condition and coil effluent composition. The empirical case evaluates its calibrated rate expression at the COT. All cases use the inputs in Table 1.

Quantity	Empirical fired	GP fired	GP electric
Single-pass ethane conversion	0.650	0.380	0.380
Single-pass ethylene yield (kg/kg coil-inlet ethane)	0.534	0.329	0.329
Fresh ethane (kg/kg ethylene)	1.218	1.150	1.150
Recycled ethane (kg/kg ethylene)	0.656	1.879	1.879
Reactor duty (GJ/t ethylene)	10.87	13.95	13.95
Electricity (MWh/t ethylene)	1.245	1.847	5.927
Total capital investment (million USD)	382	509	509
Production cost (USD/t ethylene)	423	503	696
Carbon intensity (t CO_2e/t ethylene)	1.390	1.669	2.407

The GP case requires slightly less fresh ethane because its ethylene selectivity is higher, but its lower single-pass conversion increases coil flow and recycle. Compression, refrigeration, and heating loads per tonne of ethylene increase, and the production cost with fired heating increases from 423 to 503 USD t^{-1}. The larger section loads also increase the scaled TCI from 382 to 509 million USD.

The comparison of gas temperature–time profiles holds the COT at 850 °C and residence time at 0.35 s, with the remaining inputs in Table 1. Increasing the heating exponent from 0.5 to 4 delays heating and shortens the time at temperatures near the COT within the 650–850 °C profile. Single-pass conversion decreases from 50.3% to 15.0%, and coil ethane flow increases from 2.37 to 7.66 kg kg^{-1} ethylene. More gas must be reheated, compressed, and separated to produce the same amount of ethylene, and the production cost with fired heating increases from 456 to 806 USD t^{-1} (Figure 2a–d).

We next vary COT and residence time together to map the trade-off between fresh-feed demand and downstream processing loads (**Figure 2**e,f and **Figure 3**a,b). The scan contains 5,751 conditions over 760–900 °C and 0.20–1.50 s, within both the specified operating ranges for fired and electric heating and the GP training domain. Pressure, dilution steam ratio, heating exponent, and economic inputs remain at their reference values. Production costs in this scan assume fired heating.

The maximum GP-predicted single-pass ethylene yield occurs at 900 °C and 0.906 s, and the minimum production cost occurs at 900 °C and 1.50 s. The production costs at these two conditions differ by 5.31 USD t^{-1}, or 1.42% of the production cost at maximum yield. Both conditions are at the upper COT limit of the scan, and the minimum-cost condition is also at the upper residence-time limit. Relative to the reference case, the minimum-cost condition decreases ethane recycle from 1.879 to 0.158 kg kg^{-1} ethylene and cold-box plus C_2-splitter electricity from 1,403 to 757 kWh t^{-1}.

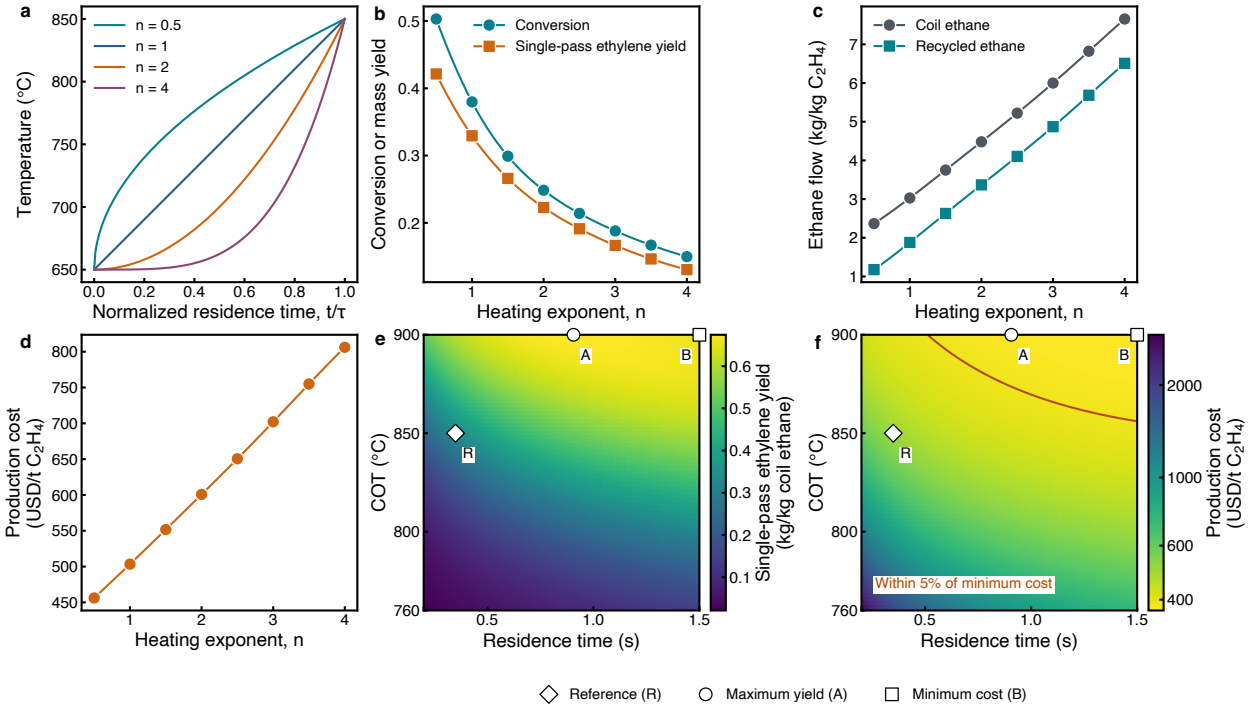


Figure 2. Heating history and operating conditions. **(a)** Gas temperature–time profiles from 650 to 850 °C with heating exponent n. **(b)** Conversion and single-pass ethylene yield, **(c)** coil and recycled ethane flows and **(d)** fired-case production cost versus n at 0.35 s. **(e)** Single-pass ethylene yield and **(f)** production cost over COT and residence time at $n = 1$. The orange contour bounds costs within 5% of the minimum. R, A and B mark the reference, maximum-yield and minimum-cost cases. Pressure is 1.5 bar and steam/ethane is 0.35 kg kg^{-1}.

Fresh ethane demand increases from 1.150 to 1.345 kg kg^{-1} ethylene as ethylene selectivity decreases. Under the stated prices and cost correlations, utility and capital savings together with changed coproduct credits outweigh the additional fresh-feed cost, and the production cost decreases from 503 to 369 USD t^{-1}.

Applying the process model to direct Cantera calculations at R, A, and B gives production costs of 500.6, 375.6, and 367.6 USD t^{-1}, respectively. The direct single-pass ethylene yields at A and B are 0.6683 and 0.6565 kg per kg coil-inlet ethane, in the same order as the GP predictions: the longer residence time at B decreases both the single-pass ethylene yield and the production cost. Increasing the number of temperature segments from 20 to 40 at B changes the production cost by 0.10 USD t^{-1}.

3.3 Break-even electricity price and grid emission factor

At the reference operating condition, electric heating uses 4.080 MWh more electricity per tonne of ethylene than fired heating. At 7.0 cents/kWh, this substitution increases the production cost by 193 USD/t ethylene. Feedstock, process electricity, capital, and other shared charges are unchanged, so the cost difference equals the difference in the cost of reactor heat.

Let q be the delivered reactor duty in GJ t^{-1} ethylene, p_e the electricity price in USD/kWh, and p_g the gas price in USD/GJ. The equations use USD/kWh, and multiplying by 100 gives the cents/kWh reported in the text and figures. The heat-delivery efficiencies are η_e for electric heating and η_f for fired heating. Because purchased natural gas and exported fuel gas have the same price per GJ (LHV), the cost difference is

$$\Delta C_E = q\left(\frac{p_e}{0.0036\eta_e} - \frac{p_g}{\eta_f}\right) \tag{4}$$

The break-even electricity price is therefore

$$p_e^* = 0.0036 p_g \frac{\eta_e}{\eta_f} \tag{5}$$

At a natural-gas price of 4 USD/GJ, electric heating gives the lower production cost below 2.28 cents/kWh for fired and electric heat-delivery efficiencies of 60% and 95%, respectively (**Figure 3**c). Reactor duty scales the cost difference but cancels from the break-even price because both heating modes deliver the same duty and value fuel gas at the same price.

At a grid emission factor of 0.36 kg CO_2e kWh^{-1}, the carbon intensity is 1.669 t CO_2e t^{-1} ethylene for fired heating and 2.407 for electric heating. Electric heating has the lower calculated carbon intensity below a break-even grid emission factor of 0.179 kg CO_2e kWh^{-1}, with eventual combustion of all fuel gas and the default coproduct credits (**Figure 3**d).

3.4 Comparison with published process cases

We compare calculated effluent composition, feed demand, and utility consumption with four published process cases (**Figure 4**). Where a source reports incomplete operating conditions, one reported quantity is used to select consistent operating conditions, and the remaining outputs are then compared. Matching the residual ethane fraction reported by Chen et al.[1] gives ethylene and hydrogen dry mole fractions whose ranges include the reported values, and methane mole fractions 1.08 to 1.71 times the reported value. For Shin et al.,[2] operating conditions selected to match fresh-feed demand give a calculated combustion heat (LHV) of 17.20–18.58 GJ t^{-1} compared with

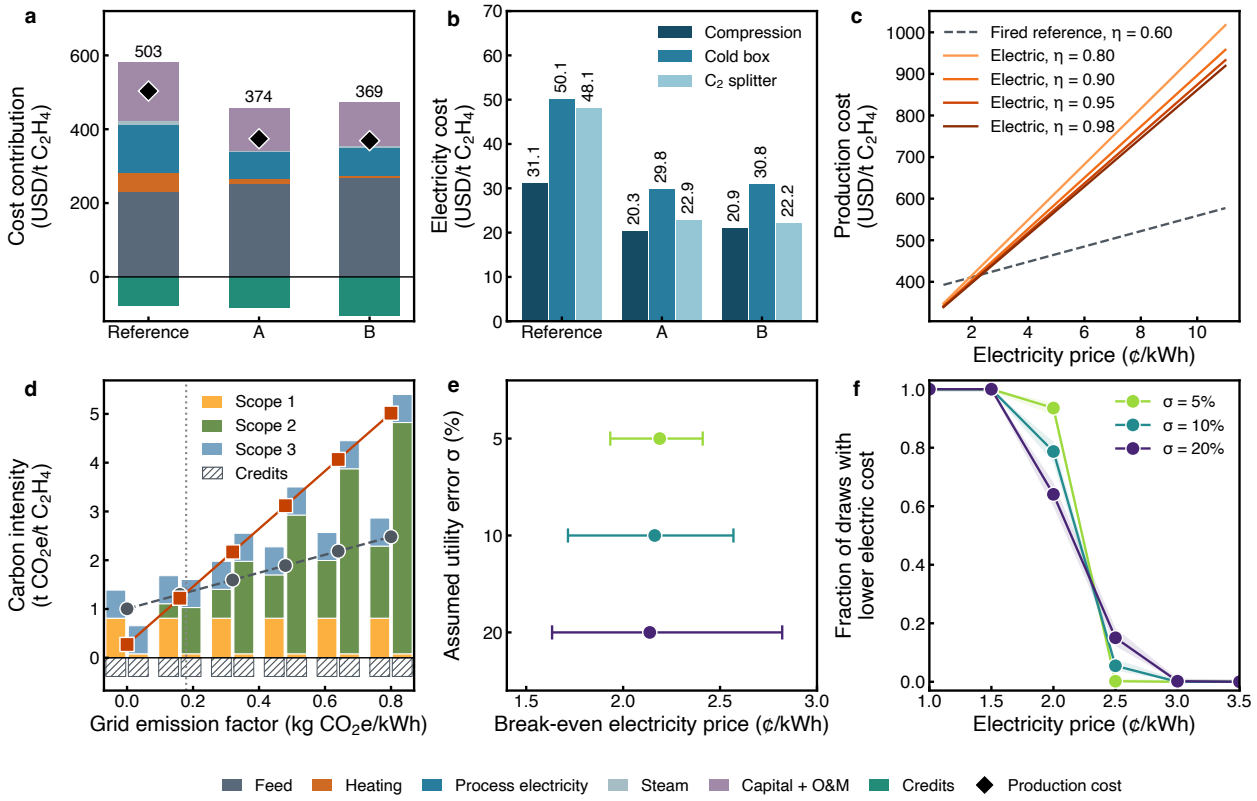


Figure 3. Cost structure, electricity price and uncertainty. **(a)** Production-cost contributions and **(b)** separation electricity costs at R, A and B (Figure 2e,f). Diamonds give net cost after credits. **(c)** Production cost versus electricity price at 4 USD GJ^{-1} gas for fired (efficiency 0.60) and electric (0.80–0.98) heating. **(d)** Carbon intensity versus grid emission factor for fired (left bars) and electric (right bars) heating by scope, with credits hatched. **(e)** Median break-even electricity price with 95% intervals for utility-discrepancy scales of 5–20%. **(f)** Fraction of 512 parameter sets with lower electric-heating cost.

17.2 GJ t^{-1} reported. The corresponding C_3+ production is 8–20% below the published quantity. For the conventional steam-cracking case of Mittal et al.,[13] matching the reported fresh ethane demand of 1.19 kg per kg ethylene at 860 °C and 1 bar requires residence times of 0.77–1.37 s for dilution steam ratios of 0.20–0.43. The calculated net specific energy consumption (SEC) is 17.4–18.4 GJ t^{-1} ethylene compared with 14.7 GJ t^{-1} reported. Process electricity is 1.09–1.25 kWh per kg ethylene compared with 0.87–1.02 kWh per kg ethylene calculated from the reported electricity expense at 6–7 cents/kWh, and C_3+ production is 2.8–3.5 times the reported LPG production (**Figure 4**e). Figure 4 distinguishes the quantities used to select these operating conditions from the compared outputs.

The Wang et al.[21] case shows a larger discrepancy in conversion. At the residence time calculated from the reported geometry and flow rate, the model predicts at most 36.6% single-pass conversion over the tested heating-exponent range, compared with 68.5% reported. The closest conversion fit is at the lower limit of the tested range, $n = 0.45$, and the calculated ethane recycle is 3.37 times the reported value (**Figure 4**c). Applying the same separation model to the published and GP-predicted effluent compositions gives electricity costs of 75.8 and 115.7 USD per tonne of final ethylene, respectively, a 53% increase (**Figure 4**d). The source case uses a different kinetic model, inlet temperature, and feed composition. SI Section S5 lists these inputs and the common separation assumptions.

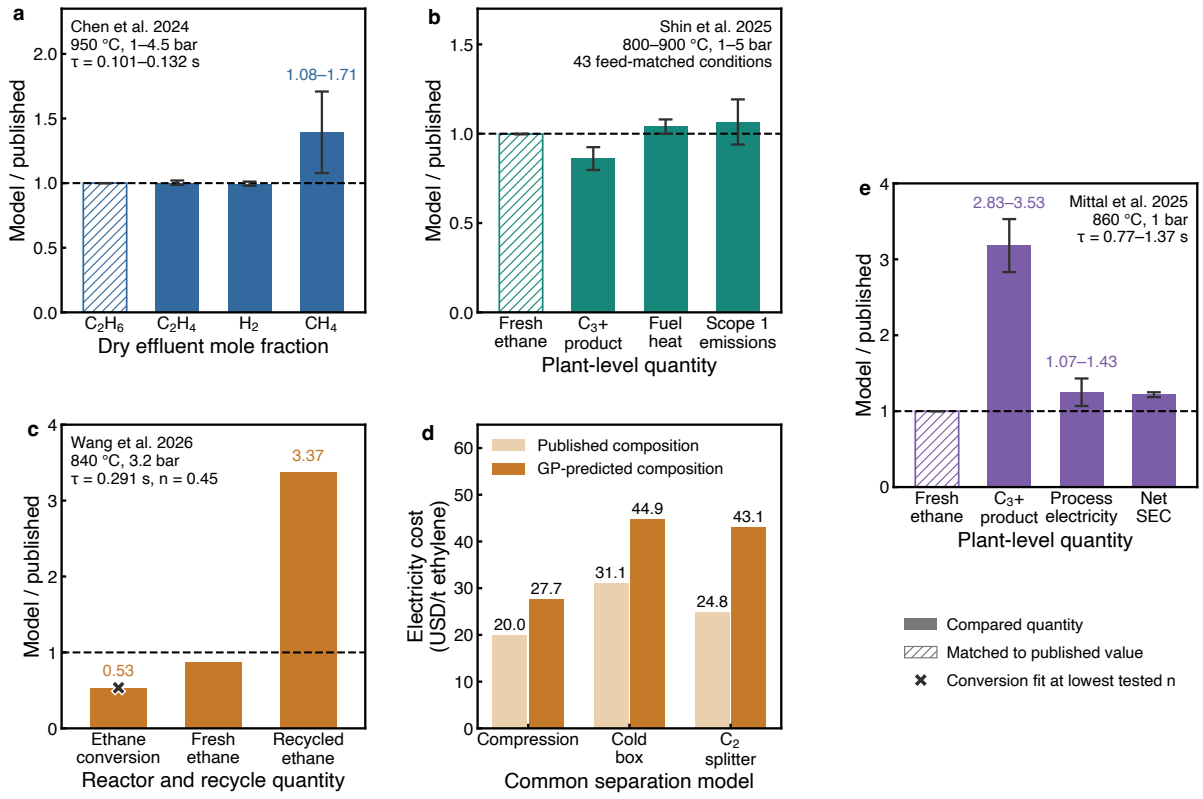


Figure 4. Literature comparisons and separation costs. **(a)** Chen et al.: dry effluent composition with residence time set by the reported residual ethane fraction at two possible reactor pressures. **(b)** Shin et al.: feed, coproduct, heat, and direct (scope 1) emissions for 43 operating conditions selected to match fresh-feed demand. **(c)** Wang et al.: single-pass conversion and fresh and recycled ethane at the residence time calculated from the reported geometry. Values in (a–c) are normalized by published quantities. Whiskers show the range of matched operating conditions, hatched bars mark quantities used to select the operating conditions, and the cross marks the closest conversion fit at $n = 0.45$, the lower limit of the tested range. **(d)** Electricity costs per tonne of final ethylene calculated from the published effluent composition (Wang et al., Table 5) and the GP-predicted (AramcoMech 3.0) composition, using identical acetylene-hydrogenation and separation models and 6.0 cents/kWh electricity (SI Section S5). **(e)** Mittal et al.: fresh ethane, C_3+ production, process electricity, and SEC for the conventional steam-cracking case at 860 °C and 1 bar, normalized by published quantities. Whiskers span dilution steam ratios of 0.20–0.43 and, for electricity, a reference taken as the reported electricity expense divided by 6–7 cents/kWh. Sources: Chen, Shin, Wang, and Mittal et al.[1,2,13,21]

3.5 Uncertainty in the break-even electricity price

Calibrating the effective fired efficiency to Shin's combustion demand changes the break-even electricity price. For the assumed 10% utility discrepancy, calibrated fired efficiency has a median of 63.75%, with a 2.5th–97.5th percentile interval of 52.75–79.00%. With 95% electric heating efficiency and equal furnace capital factors, the median break-even electricity price is 2.16 cents/kWh and its 2.5th–97.5th percentile interval is 1.72–2.57 cents/kWh. Increasing the assumed discrepancy from 5% to 20% widens this interval from 1.93–2.41 to 1.63–2.82 cents/kWh (**Figure 3**e).

At 2.0 cents/kWh, electric heating gives the lower production cost in 93.6%, 78.7%, and 64.1% of sampled parameter sets for the 5%, 10%, and 20% discrepancy cases, respectively (**Figure 3**f). At the reference electricity price of 7.0 cents/kWh, the median increase in production cost with electric heating is 196–198 USD t^{-1}. Equation (5) shows that, at the same coil effluent composition, coil duty, and furnace capital factors, shared process costs cancel and the break-even electricity price depends on fired efficiency. Discrepancies in Shin's C_3+ production and Chen's methane fraction and fresh-feed demand remain after utility calibration.

3.6 Replacing reactor and process submodels

Because the gas temperature–time profile is an input, a heat-transfer model of a specific reactor configuration can provide it. Volume-averaged structured-reactor models, for example, retain temperature-dependent transport and external heat loss at lower computational cost than channel-resolved calculations[22].

The scenario files record all inputs, so each comparison can be repeated. Changing the kinetic mechanism requires new yield and enthalpy calculations. Separation correlations can be revised with the kinetic surrogate held fixed, and prices and the grid emission factor can be changed with the material and energy balance held fixed.

Cassady et al. measured species time histories during dilute ethane/argon pyrolysis and found temperature-dependent differences among kinetic mechanisms, including underprediction of lower-temperature ethane decomposition by AramcoMech 3.0[23]. The run-length correlation is specified independently of the gas-phase mechanism, so users can revise availability as experimental coking and decoking data become available.

4 Conclusions

Delaying heating at fixed COT and residence time increases ethane recycle and the associated heating and separation loads per tonne of ethylene. In the COT–residence-time scan, maximum single-pass ethylene yield and minimum production cost occur at different conditions because lower heating cost and higher coproduct credits outweigh additional feed and processing costs. Comparisons with published process cases quantify how differences in calculated effluent composition change downstream electricity costs. At fixed coil effluent composition, coil duty, and equipment costs, energy prices and heat-delivery efficiencies set the break-even electricity price between fired and electric heating.

Code and data availability

The model, training and test data, and release records are available at https://github.com/robin-yk/TEA-EthaneCracking. The companion manuscript files contain the scenario outputs,

literature comparisons, conditional uncertainty results, reproduction checks, figure scripts, and this TeX source.

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Supporting Information

S1. Reactor equations, surrogate, and numerical checks

S1.1. Constant-pressure species balances

The kinetic calculation treats a closed fluid element at prescribed temperature and constant pressure. Let Y_k denote species mass fraction, W_k molecular weight in kg kmol^{-1}, ρ mixture density, and $\dot{\omega}_k$ net molar production rate in kmol m^{-3} s^{-1}. The integrated species equations are

$$\begin{aligned} \frac{dY_k}{dt} &= \frac{W_k \dot{\omega}_k}{\rho} \\ \dot{\omega}_k &= \sum_r (\nu''_{kr} - \nu'_{kr}) \mathcal{R}_r \\ \rho &= \frac{P \overline{W}}{R_u T} \\ \overline{W}^{-1} &= \sum_k \frac{Y_k}{W_k} \end{aligned} \tag{S1}$$

Here ν'_{kr} and ν''_{kr} are reactant and product stoichiometric coefficients, $\mathcal{R}_r$ is the net rate of reaction r, and R_u is the gas constant on a kmol basis. Cantera evaluates these rates from the published mechanism, including its pressure-dependent and third-body expressions. For an elementary mass-action reaction, the rate has the form $\mathcal{R}_r = k_{f,r} \prod_k c_k^{\nu'_{kr}} - k_{b,r} \prod_k c_k^{\nu''_{kr}}$, where $c_k = \rho Y_k / W_k$. The reaction-specific mechanism definitions determine the actual rate law. The energy balance is not solved because Eq. (1) specifies temperature.

For N_s temperature segments, the segment times and temperatures are

$$\begin{aligned} u_i &= \frac{i}{N_s} \\ f_i &= u_i^{1/n} \\ \Delta t_i &= \tau (f_{i+1} - f_i) \\ T_i &= T_{\text{in}} + (T_{\text{out}} - T_{\text{in}}) \left(\frac{f_i + f_{i+1}}{2} \right)^n \end{aligned} \tag{S2}$$

for $i = 0, \ldots, N_s - 1$. The final composition of each segment is the initial composition of the next, and each segment uses constant T_i and P. The calculations use $N_s = 20$, with relative and absolute integration tolerances 10^{-8} and 10^{-15}. At the final composition, the temperature is set to T_{out} to evaluate the outlet enthalpy without an additional reaction interval.

The inlet basis is $n_{A,0} = 1$ kmol ethane, where A denotes ethane and E denotes ethylene. With dilution steam ratio d (kg steam per kg ethane), inlet ethane mass is $m_{A,0} = W_A n_{A,0}$, steam mass

is $dm_{A,0}$, and total mass is $m_0 = (1+d)m_{A,0}$. Outlet amounts and reactor outputs are

$$\begin{aligned} n_{k,\mathrm{out}} &= \frac{m_0 Y_{k,\mathrm{out}}}{W_k} \\ X &= \frac{\max(0, n_{A,0} - n_{A,\mathrm{out}})}{n_{A,0}} \\ S_E &= \frac{n_{E,\mathrm{out}}}{X n_{A,0}} \\ Y_E &= \frac{m_0 Y_{E,\mathrm{out}}}{m_{A,0}} \\ h_{\mathrm{coil}} &= \frac{m_0(h_{\mathrm{out}} - h_{\mathrm{in}})}{10^6 m_{A,0}} \end{aligned} \tag{S3}$$

The specific enthalpies h_{in} and h_{out} are in J kg^{-1} mixture, so h_{coil} is in MJ kg^{-1} inlet ethane. S_E is set to 0 when consumed ethane is at most 10^{-12} kmol. Individual species yields use the same mass denominator as Y_E. C_3 and C_{4+} sum masses of species containing exactly three and at least four carbon atoms, respectively. For element a, with b_{ak} atoms per molecule, closure is checked using

$$\begin{aligned} \epsilon_a &= \frac{\sum_k b_{ak} n_{k,\mathrm{out}} - \sum_k b_{ak} n_{k,0}}{\max(\sum_k b_{ak} n_{k,0}, 10^{-30})} \\ \epsilon_m &= \sum_k Y_{k,\mathrm{out}} - 1 \end{aligned} \tag{S4}$$

For an oxygen-free inlet, the oxygen closure error is the outlet oxygen amount itself.

S1.2. Input and output transformations

Let $\mathbf{v} = (T_{\mathrm{out}}, \log_{10}\tau, d, P, n)$, with temperature in °C, time in seconds, and pressure in bar. For input coordinate j, normalization uses the observed training extrema a_j and b_j:

$$x_j = \frac{v_j - a_j}{\max(b_j - a_j, 10^{-12})} \tag{S5}$$

For output coordinate o and training row i, the transformations are

$$z_{io} = g_o(y_{io}) = \begin{cases} \log[p_{io}/(1-p_{io})], & p_{io} = \min(1-10^{-7}, \max(10^{-7}, y_{io})) \\ \log[\max(0, y_{io}) + 10^{-10}], & \text{yield or enthalpy} \end{cases} \tag{S6}$$

The first branch applies to conversion and molar selectivity. The transformed training outputs are centered and scaled as $\tilde{z}_{io} = (z_{io} - \mu_o)/s_o$, where μ_o and s_o are the mean and population standard deviation over all 640 training rows. If $s_o < 10^{-12}$, s_o is set to 1. The same normalization constants are used for the internal fitting and tuning subsets.

S1.3. Shared covariance, fitting, and prediction

The RBF covariance uses normalized inputs:

$$k(\mathbf{x}, \mathbf{x}') = \exp\left[-\frac{1}{2}\sum_j \left(\frac{x_j - x'_j}{\ell_j}\right)^2\right] \tag{S7}$$

The selected length scales are 0.418, 0.380, 0.532, 0.570, and 0.532 for temperature, logarithmic residence time, dilution steam ratio, pressure, and heating exponent, respectively. Training and test-set sampling seeds are 42 and 4242. Hyperparameter tuning uses seed 2026. The 39 candidates comprise seven common rescalings of the initial vector $(0.22, 0.20, 0.28, 0.30, 0.28)$ by $(0.45, 0.65, 0.85, 1, 1.2, 1.5, 1.9)$ and 32 independent log-uniform five-coordinate draws between 0.055 and 1.35. All candidate coordinates are restricted to 0.035–2.5.

For a fitting set with covariance matrix $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$, the diagonal regularizer is $\lambda = 2 \times 10^{-6}$. Cholesky factorization and prediction use

$$\begin{aligned} A &= K + \lambda I = LL^\top \\ \alpha_o &= L^{-\top} L^{-1} \tilde{\mathbf{z}}_o \\ \hat{z}_o(\mathbf{x}_*) &= \mu_o + s_o \mathbf{k}_*^\top \alpha_o \\ v_* &= \max(0, 1 - \mathbf{k}_*^\top A^{-1} \mathbf{k}_*) \end{aligned} \tag{S8}$$

Here $\mathbf{k}_*$ contains covariances between the prediction point and the fitting points. The point prediction is $\hat{y}_o = g_o^{-1}(\hat{z}_o)$, with inverse maps $g_o^{-1}(z) = 1/(1 + e^{-z})$ for conversion and selectivity, and $g_o^{-1}(z) = \max(0, e^z - 10^{-10})$ for yields and enthalpy. This is the inverse-transformed latent mean. Nonlinear inverse transformation does not in general give the mean in original output units. Outputs share A but have separate coefficient vectors, with no fitted cross-output covariance.

The candidate objective is the mean over the ten outputs of tuning-set RMSE divided by that output's tuning-set range. A seeded permutation divides the 640 training points into 512 fitting and 128 tuning points. Kernel search uses a seeded 256-point subset of the fitting set. After selection, a 512-point fit gives the tuning residuals and variances used for uncertainty calibration. The final coefficients and inverse covariance use all 640 training points. The surrogate also calculates a first-order, calibrated uncertainty estimate

$$\begin{aligned} \sigma_{o,0}(\mathbf{x}_*) &= s_o \sqrt{v_*}\, d_o(\hat{z}_o) \\ d_o(z) &= \begin{cases} p(1-p), & p = (1 + e^{-z})^{-1} \\ e^z, & \text{log-transformed output} \end{cases} \\ c_o &= \mathrm{clip}_{[0.25,20]} \left[\sqrt{\frac{1}{128} \sum_{i \in \mathcal{T}} \left(\frac{\hat{y}_{io} - y_{io}}{\max(\sigma_{io,0}, 10^{-14})} \right)^2 } \right] \\ \sigma_o &= c_o \sigma_{o,0} \end{aligned} \tag{S9}$$

The set $\mathcal{T}$ is the 128-point internal tuning subset, and its residuals and uncalibrated uncertainties use the 512-point fit. The nominal interval is $\hat{y}_o \pm 1.96\sigma_o$. The variance excludes an additional observation-noise term. Process calculations use GP point predictions, and Section S1.4 evaluates the effect of uncertainty in effective process parameters.

For the $N_h = 128$ points of the independent test set, the reported statistics are

$$\begin{aligned} \mathrm{RMSE}_o &= \sqrt{N_h^{-1} \sum_i (\hat{y}_{io} - y_{io})^2} \\ \mathrm{NRMSE}_o &= \frac{\mathrm{RMSE}_o}{\max_i y_{io} - \min_i y_{io}} \\ R_o^2 &= 1 - \frac{\sum_i (\hat{y}_{io} - y_{io})^2}{\sum_i (y_{io} - \overline{y}_o)^2} \\ \mathrm{coverage}_o &= N_h^{-1} \sum_i \mathbf{1}\{|\hat{y}_{io} - y_{io}| \le 1.96 \sigma_{io}\} \end{aligned} \tag{S10}$$

Here $\overline{y}_o$ is the test-set mean, and $\mathbf{1}\{\cdot\}$ equals 1 when the condition holds and 0 otherwise.

Table S1. Surrogate accuracy on the independent test set. RMSE is normalized by the observed range of each output. Coverage is the fraction of 128 points inside the nominal 95% interval.

Output	R^2	RMSE/range (%)	Coverage (%)
Ethane conversion	0.9997	0.59	96.9
Ethylene molar selectivity	0.9993	0.39	93.0
Ethylene mass yield	0.9994	0.85	92.2
Methane mass yield	0.9996	0.26	92.2
Hydrogen mass yield	0.9995	0.76	90.6
Acetylene mass yield	0.9956	1.24	95.3
Ethane mass yield	0.9986	1.27	95.3
C_3 mass yield	0.9971	1.62	83.6
C_{4+} mass yield	0.9962	0.98	86.7
Enthalpy rise	0.9993	0.77	86.7

The maximum closure errors of the Cantera calculations are 1.28×10^{-11} for carbon, 5.69×10^{-11} for hydrogen, 1.26×10^{-11} for oxygen, and 2.00×10^{-15} for mass. The segment-test acceptance thresholds are 0.005 absolute error for conversion, selectivity, and ethylene mass yield, and 1% relative error for enthalpy.

With the final transforms and kernel held fixed, nested farthest-point training subsets of 64, 128, 256, 512, and 640 points give mean range-normalized errors of 6.19%, 3.63%, 2.40%, 1.13%, and 0.87%, respectively, on the same 128-point independent test set. This test measures the effect of the number of training points within the training domain.

The GP output transformations keep yields positive and conversion and selectivity between 0 and 1. They do not constrain elemental closure across outputs. For example, one predicted ethane yield in the test set is 1.023 kg per kg coil-inlet ethane, whereas the corresponding Cantera value is below 1. The process recycle calculation uses conversion and the ethane formed by acetylene hydrogenation, as shown in Eq. (2). The GP-predicted ethane yield is used in the C_2 flow that sets the acetylene target of the acetylene hydrogenation reactor. The scenario files include diagnostic checks. The conversion check uses the 0.45–0.85 conversion range of the empirical rate model and therefore flags the reference case even though the GP inputs lie inside the kinetic training domain. This check is separate from the GP domain check.

Kim et al. incorporate differential-equation residual penalties into GP marginal-likelihood hyperparameter estimation[24]. Adding element-balance residuals of the predicted yields to the present tuning objective is a possible extension. The present model selects hyperparameters by prediction error alone.

S1.4. Conditional utility calibration and paired uncertainty

Chen's reported compressor power, 29.4 MW, divided by ethylene production, 79.4 t h^{-1}, sets the compression observation. Residence time is first fitted to the reported dry ethane mole fraction at either 1 or 4.5 bar, and the two pressure interpretations receive equal prior weight. Shin's 17.2 GJ t^{-1} total combustion energy sets the heat observation. The corresponding operating-condition scan covers 800–900 °C in 5 °C steps and 1–5 bar in 0.5 bar steps, with residence time in 0.02–1.5 s fitted to fresh-feed demand. Operating conditions within 0.5% of that demand receive

equal prior weight. The reported residual-ethane fraction and fresh-feed demand are treated as fixed inputs when selecting these conditions.

Let z_j denote a selected operating condition, d_o the utility observation, and $F_o(\theta_i, z_j)$ the corresponding model output at parameter node θ_i. A uniform prior and Gaussian aggregate mismatch give normalized quadrature weights

$$\begin{aligned} \widetilde{w}_{ij} &= a_i \exp\left[-\frac{(F_o(\theta_i, z_j) - d_o)^2}{2\sigma_o^2}\right] \\ w_{ij} &= \frac{\widetilde{w}_{ij}}{\sum_{i',j'} \widetilde{w}_{i'j'}} \\ \sigma_o &= \delta d_o \\ \delta &\in \{0.05, 0.10, 0.20\} \end{aligned} \tag{S11}$$

Each parameter grid has 161 evenly spaced nodes, with trapezoidal weights $a_i = 1/2$ at its endpoints and 1 elsewhere. Fired efficiency has prior $\eta_f \sim U(0.45, 0.85)$, and the compression multiplier has prior $k_c \sim U(0.5, 2.0)$. Their outputs are $F_{\text{heat}} = q_{\text{demand}}(z)/\eta_f$ and $F_{\text{comp}} = k_c w_c(z)$ in units matching each observation. The two studies are treated as conditionally independent. These parameters account for differences in equipment and process boundaries. The mismatch scales and prior limits are assumed. Other species and emission outputs from the same studies are excluded from the calibration and used for validation. Their reported intervals contain parameter variation without added mismatch noise. Financial quantities and Wang's different reactor boundary are excluded from the likelihood.

Each scenario draws 512 parameter pairs using seed 20260923. The fitted η_f and k_c are sampled from their marginal quadrature weights. Heat-recovery fraction $r_h \sim U(0.5, 0.9)$ and shared capital multiplier $k_I \sim U(0.7, 1.5)$ are sampled independently from their priors. Electric efficiency is fixed at 0.90, 0.95, or 0.98, and its furnace-only capital multiplier at 0.7, 1.0, or 1.3. Together with three mismatch scales, these choices define 27 scenarios. Both heating modes use the same draw of k_c, r_h, k_I and the same reference chemistry and prices. The calculation holds GP outputs at their point predictions.

For parameter set b, cost varies linearly with electricity price p_e. Evaluations at 0 and 10 cents/kWh (0 and 0.1 USD/kWh in the equations) determine

$$\begin{aligned} A_b &= C_{e,b}(0) - C_{f,b}(0) \\ B_b &= \frac{C_{e,b}(0.1) - C_{f,b}(0.1) - A_b}{0.1} \\ \Delta C_b(p_e) &= A_b + B_b p_e \\ p^*_{e,b} &= -A_b/B_b \end{aligned} \tag{S12}$$

Here e and f denote electric and fired heating. Quantiles at 2.5%, 50%, and 97.5% summarize production costs, paired differences, and break-even electricity prices. The fraction with $\Delta C_b < 0$ estimates the conditional frequency of a lower production cost with electric heating, with a Wilson interval for Monte Carlo counting uncertainty. Common process charges cancel in the difference when furnace capital factors are equal, and the break-even price then reduces to Eq. (5). A furnace capital factor specific to one heating mode adds a capital-cost difference.

At 10% assumed mismatch, the fitted fired-efficiency interval is 52.75–79.00% and the compression-multiplier interval is 0.988–1.475. Shin's C_3+ production, 0.1376 kg kg^{-1} ethylene, remains above the conditional interval of 0.1114–0.1270. Chen's methane fraction, fresh-feed demand, and reactor heat also remain outside their parameter intervals.

S2. Implemented process correlations

S2.1. Material and heat balances

All process masses below are kg per kg final ethylene, and all process duties and work are kJ per kg final ethylene. Let y_k^{coil} be a GP species yield on the inlet-ethane basis and W_k its molecular weight in g mol^{-1}. The balances of the acetylene hydrogenation reactor are written on this basis before the flows are scaled to the ethylene basis. With A denoting ethane and E ethylene, its molar balances are

$$\begin{aligned}
\nu_k &= 1000 y_k^{\mathrm{coil}}/W_k \\
\nu_{\mathrm{C_2}} &= \nu_E + \nu_A + \nu_{\mathrm{C_2H_2}} \\
\nu_{\mathrm{res}} &= \min(\nu_{\mathrm{C_2H_2}}, 5\times10^{-6}\nu_{\mathrm{C_2}}) \\
\xi &= \nu_{\mathrm{C_2H_2}} - \nu_{\mathrm{res}} \\
\Delta Y_E &= 0.90\xi W_E/1000 \\
\Delta Y_A &= 0.10\xi W_A/1000 \\
Y_{\mathrm{H_2,need}} &= 1.10\xi W_{\mathrm{H_2}}/1000
\end{aligned} \tag{S13}$$

The quantities ν_k and ξ are mol per kg coil-feed ethane. The 5 ppm target and 90% ethylene selectivity are fixed assumptions. Available process hydrogen is consumed first. Any shortfall is recorded as external hydrogen, so hydrogen availability does not reduce the imposed acetylene conversion. External hydrogen purchase and the heat of the hydrogenation reaction are omitted from the cost and heat balances, respectively. The capital cost of the acetylene hydrogenation reactor is included in the recovery-section reference cost. Masses after acetylene hydrogenation are

$$\begin{aligned}
m_{\mathrm{H_2}} &= \max(0, y_{\mathrm{H_2}}^{\mathrm{coil}} - Y_{\mathrm{H_2,need}}) m_{\mathrm{coil}} \\
m_{\mathrm{C_2H_2}} &= (\nu_{\mathrm{res}} W_{\mathrm{C_2H_2}}/1000) m_{\mathrm{coil}} \\
m_k &= y_k^{\mathrm{coil}} m_{\mathrm{coil}} \quad (k = \mathrm{CH_4}, \mathrm{C_3}, \mathrm{C_{4+}}) \\
m_s &= d\, m_{\mathrm{coil}} \\
S_{E,\mathrm{process}} &= \frac{Y_E W_A}{\max(X, 10^{-12}) W_E} \\
S_{E,\mathrm{mass}} &= \frac{Y_E}{\max(X, 10^{-12})}
\end{aligned} \tag{S14}$$

Here m_s is dilution steam and d is the dilution steam ratio. $S_{E,\mathrm{mass}}$ is the ultimate ethylene yield (kg ethylene per kg ethane consumed). Solid-coke mass m_C is zero for the GP chemistry. A difference exceeding 0.01 between derived and independently predicted molar selectivity is flagged in the diagnostic checks. The 0.995 separation-recovery specification is used in the cold-box correlation below. The product-basis mass balance has no additional recovery-loss multiplier.

Upstream preheat uses heat capacities of 3.0 and 2.3 kJ kg^{-1} K^{-1} for ethane and steam. Total sensible heat is calculated from 150 °C to the COT. The TLE credit is 0.70 times this sensible duty. Dilution-steam generation uses 2100 kJ kg^{-1} and is charged at $0.45 p_{\mathrm{steam}} + 0.40$ USD GJ^{-1}, where

the latter term represents cooling-water rejection.

$$\begin{aligned} C_p^* &= 3.0m_{\text{coil}} + 2.3m_s \\ q_{\text{sens}} &= C_p^*(T_{\text{out}} - 150) \\ q_{\text{demand}} &= C_p^*(650 - 150) + 1000m_{\text{coil}}h_{\text{coil}} \\ q_{\text{rxn}} &= q_{\text{demand}} - q_{\text{sens}} \\ q_{\text{input}} &= q_{\text{demand}}/\eta_h \\ q_{\text{TLE}} &= 0.70q_{\text{sens}} \\ q_{\text{DSG}} &= 2100m_s \end{aligned} \tag{S15}$$

Temperature differences use °C or K interchangeably, and η_h is heat-delivery efficiency. Without heat recovery, $q_{\text{TLE}} = 0$. The quantity q_{rxn} is the residual after the constant-heat-capacity sensible partition. The GP total duty uses the mechanism enthalpy directly.

The fuel-gas energy (LHV) and heating-mode balances are

$$\begin{aligned} q_f &= 1000(50m_{\text{CH}_4} + 120m_{\text{H}_2}) \\ (q_{\text{buy}}, q_{\text{sell}}, w_h)_{\text{fired}} &= (\max(0, q_{\text{input}} - q_f), \max(0, q_f - q_{\text{input}}), 0) \\ (q_{\text{buy}}, q_{\text{sell}}, w_h)_{\text{electric}} &= (0, q_f, q_{\text{input}}) \end{aligned} \tag{S16}$$

The methane and hydrogen lower heating values are 50 and 120 MJ kg^{-1}. The symbols q_{buy} and q_{sell} denote purchased natural gas and exported fuel gas (LHV), and w_h denotes electric heating input.

S2.2. Compression and refrigeration

The compressor feed is the dry cracked gas before acetylene hydrogenation. On the final-product basis, its ethylene and ethane masses are $m_E^{\text{raw}} = \max(0, 1 - \Delta Y_E m_{\text{coil}})$ and $m_A^{\text{raw}} = \max(0, m_{\text{recycle}} - \Delta Y_A m_{\text{coil}})$. Its hydrogen and acetylene masses are $m_k^{\text{raw}} = y_k^{\text{coil}} m_{\text{coil}}$. Methane, C_3, and C_{4+} masses remain unchanged. The total molar amount is $\mathcal{N}_g^{\text{raw}} = \sum_k 1000 m_k^{\text{raw}}/W_k$. The molecular weights of ethylene, ethane, methane, hydrogen, acetylene, C_3, and C_{4+} are 28.05, 30.07, 16.04, 2.016, 26.038, 42.08, and 56.11 g mol^{-1}, respectively, and the last two represent lumped products.

Compression uses four equal pressure-ratio stages with complete intercooling to 313 K, heat-capacity ratio $\gamma = 1.32$, and efficiency $\eta_c = 0.75$:

$$\begin{aligned} r_s &= \left(\frac{32}{1.5}\right)^{1/4} \\ w_c &= \frac{4\gamma R(313)\mathcal{N}_g^{\text{raw}}}{1000(\gamma - 1)\eta_c}\left(r_s^{(\gamma-1)/\gamma} - 1\right) \end{aligned} \tag{S17}$$

Here $R = 8.314$ J mol^{-1} K^{-1}. The suction pressure remains 1.5 bar even if the reactor-pressure input changes.

The cold-box calculation uses the amounts after acetylene hydrogenation, $\mathcal{N}_k = 1000m_k/W_k$, with $m_E = 1$ and $m_A = m_{\text{recycle}}$. The total molar amount is $\mathcal{N}_g = \sum_k \mathcal{N}_k$ and the total mass is $m_g = \sum_k m_k$. The fuel-gas (methane plus hydrogen) mole fraction is $\phi = (\mathcal{N}_{\text{CH}_4} + \mathcal{N}_{\text{H}_2})/\mathcal{N}_g$.

With specified recovery $r_E = 0.995$, the cold-end temperature t_c in °C and cold-box duty are

$$\begin{aligned} t_c &= -100 - 40\log_{10}\left(\frac{0.01}{1-r_E}\right) - 160(\phi - 0.40) \\ q_c &= 2.2 m_g (30 - t_c) + 350(1 + m_{\text{recycle}} + m_{\text{C}_2\text{H}_2} + m_{\text{C}_3} + m_{\text{C}_{4+}}) \\ w_r &= q_c \sum_{j=1}^{3} \frac{a_j}{\max[0.05, \eta_2 T_j/(T_a - T_j)]} \end{aligned} \tag{S18}$$

The assumed sensible and latent coefficients are 2.2 kJ kg^{-1} K^{-1} and 350 kJ kg^{-1}, respectively. The duty fractions are $\mathbf{a} = (0.55, 0.30, 0.15)$ at $\mathbf{T} = (233.15, 173.15, t_c{+}273.15)$ K. Ambient temperature is $T_a = 303$ K and second-law efficiency is $\eta_2 = 0.35$. The denominator is a Carnot coefficient of performance multiplied by η_2, with the implemented numerical floor of 0.05. The temperature and duty relations are screening correlations.

S2.3. C_2 splitter shortcut

For discharge pressure $P_d = 32$ bar, relative volatility $\alpha_v = 1.85 - 0.018 P_d$ and distillate ethylene mole fraction $x_D = 0.999$ give

$$\begin{aligned} N_{\min} &= \frac{\log[(x_D/(1-x_D))(0.99/0.01)]}{\log \alpha_v} \\ N &= \text{round}(2.1 N_{\min}) \\ z_F &= \min\left(0.995, \frac{\mathcal{N}_E}{\mathcal{N}_E + \mathcal{N}_A}\right) \end{aligned} \tag{S19}$$

The Fenske expression assumes a 0.01 ethylene mole fraction in the bottoms. The capped binary-feed mole fraction z_F is used in the shortcut reflux calculation:

$$\begin{aligned} R_{\min} &= \max\left[0.5, \frac{x_D/z_F - \alpha_v(1-x_D)/(1-z_F)}{\alpha_v - 1}\right] \\ g_N &= \frac{N - N_{\min}}{N+1} \\ R_s &= \max\left[1.15 R_{\min}, 1.05\{R_{\min} + 0.9(1 - \sqrt{g_N})\}\right] \\ w_s &= \frac{280(R_s+1)}{\max[0.05, \eta_2(243)/(T_a - 243)]} \end{aligned} \tag{S20}$$

Here R_s is the operating reflux ratio and w_s is splitter condenser refrigeration work. The 280 kJ kg^{-1} latent-duty coefficient and 243 K condenser temperature are assumed. This shortcut calculates the splitter utility and capital loads.

S2.4. Capital, availability, and production cost

The scaled bare-module cost, in million USD, is

$$\begin{aligned} B = 126.8 \left(\frac{K}{610}\right)^{0.65} \Bigg[& 0.30 \left(\frac{m_{\text{coil}}}{1.809}\right)^{0.65} + 0.25 \left(\frac{w_c}{1134}\right)^{0.70} \\ & + 0.25 \left(\frac{w_r}{2098}\right)^{0.70} + 0.20 \left(\frac{R_s N}{195}\right)^{0.60} \Bigg] \end{aligned} \tag{S21}$$

Here K is nominal capacity in kt y^{-1}, w_c and w_r are compression and refrigeration work in kJ kg^{-1} ethylene, R_s is reflux ratio, and N is the actual stage count. The TCI is $2.5B$. Annual nonlabor operation and maintenance are 7.5% of the TCI.

The run-length correlation, in days, is

$$d_{\mathrm{run}} = 60\exp[-0.025(T_{\mathrm{out}} - 850)]\left(\frac{d}{0.35}\right)^{0.6}\left(\frac{0.35}{\tau}\right)^{0.3} \tag{S22}$$

The on-stream factor is $d_{\mathrm{run}}/(d_{\mathrm{run}}+1.5)$, with an assumed 1.5-day shutdown between runs. It equals 0.9756 in all three reference cases and throughout the heating-exponent sweep. This availability correlation is an assumed process input. The GP kinetic model includes only gas-phase reactions, and the process model sets the solid-coke yield to zero for the GP chemistry.

Let F_{on} be this on-stream factor, $M = 1000KF_{\mathrm{on}}$ annual production in t y^{-1}, $I = 2.5B \times 10^6$ the TCI in USD, f_c annual capital charge fraction, and L_a annual labor cost in USD y^{-1}. With electricity price p_e in USD kWh^{-1}, gas and steam prices p_g, p_s in USD GJ^{-1}, and ethane and C$_3$+ prices p_A, p_L in USD t^{-1}, all cost terms in USD t^{-1} ethylene are

$$\begin{aligned}
C_{\mathrm{feed}} &= m_{\mathrm{fresh}}p_A \\
C_{\mathrm{heat}} &= 10^{-3}q_{\mathrm{buy}}p_g + \frac{w_h}{3.6}p_e \\
C_{\mathrm{process\ electricity}} &= \frac{w_c + w_r + w_s}{3.6}p_e \\
C_{\mathrm{dilution}} &= 10^{-3}q_{\mathrm{DSG}}(0.45p_s + 0.40) \\
C_{\mathrm{capital}} &= \frac{f_c I}{M} \\
C_{\mathrm{O\&M}} &= \frac{0.075I + L_a}{M} \\
C_{\mathrm{credits}} &= 10^{-3}(q_{\mathrm{TLE}}p_s + q_{\mathrm{sell}}p_g) + (m_{\mathrm{C_3}} + m_{\mathrm{C_{4+}}})p_L
\end{aligned} \tag{S23}$$

Equation (3) combines these terms. The factor 10^{-3} converts kJ kg^{-1} to GJ t^{-1}, and division by 3.6 converts the same duty to kWh t^{-1}. In the figures, the feed cost is divided into $C_{\mathrm{floor}} = (W_A/W_E)p_A$ and $C_{\mathrm{selectivity\ loss}} = [m_{\mathrm{fresh}} - W_A/W_E]p_A$, and their sum is C_{feed}. Total electricity demand is $E_t = (w_c + w_r + w_s + w_h)/3.6$ kWh t^{-1}, and the SEC is $10^{-3}(q_{\mathrm{input}} - q_{\mathrm{TLE}} + w_c + w_r + w_s)$ GJ t^{-1} ethylene.

S2.5. Empirical reference chemistry

The empirical rate model uses first-order ethane conversion kinetics with an activation energy of 272.8 kJ mol^{-1} and a pre-exponential factor chosen to give $X = 0.65$ at 850 °C and 0.35 s. Its selectivity is the quadratic in X through the points $(X, S) = (0.50, 0.93)$, $(0.65, 0.88)$, and $(0.80, 0.82)$, with added dilution and residence-time corrections. These points are assumed reference values and define the empirical reference case in Table 2.

$$\begin{aligned}
A_0 &= \frac{-\log(1-0.65)}{0.35}\exp\left[\frac{272800}{8.314(850+273.15)}\right] \\
X &= \max\{10^{-4}, 1 - \exp[-A_0 e^{-272800/[8.314(T_{\mathrm{out}}+273.15)]}\tau]\} \\
S &= \mathrm{clip}_{[0.55,0.985]}\left[\sum_{i=1}^{3} S_i \prod_{j\neq i}\frac{X - X_j}{X_i - X_j} + 0.06(d - 0.35) + 0.04\log(0.35/\tau)\right]
\end{aligned} \tag{S24}$$

For lost-selectivity fraction $\ell = 1-S$, the assumed fractions of the lost carbon converted to methane, coke, and C_3 are

$$\begin{aligned} f_M &= \min(0.80, 0.45 + 0.35\ell/0.30) \\ f_C &= \min(0.25, 0.08 + 0.15\ell/0.30) \\ f_3 &= \max(0.02, 1 - f_M - f_C) \end{aligned}$$

Per mole of consumed ethane, product amounts are

$$\begin{aligned} b_M &= 2\ell f_M \\ b_C &= 2\ell f_C \\ b_3 &= 2\ell f_3/3 \\ b_H &= 3 - 2S - 2\ell(2f_M + f_3) \end{aligned} \tag{S25}$$

The required consumed ethane is $\mathcal{N}_A^{\text{consumed}} = 1000/(SW_E)$ mol per kg ethylene. The resulting masses are

$$\begin{aligned} m_{\text{fresh}} &= \mathcal{N}_A^{\text{consumed}} W_A/1000 \\ m_k &= \mathcal{N}_A^{\text{consumed}} b_k W_k/1000 \\ m_{\text{coil}} &= m_{\text{fresh}}/X \end{aligned}$$

The empirical C_3 lump uses propylene stoichiometry and $W_C = 12.011$ g mol^{-1} for coke. Acetylene and C_{4+} are zero. The empirical heat duty is $q_{\text{demand}} = 145\mathcal{N}_A^{\text{consumed}} + q_{\text{sens}}$, using an assumed 145 kJ mol^{-1} reaction heat. The utility, capital, and cost equations are the same as for the GP chemistry.

S3. Carbon accounting for the plotted scenarios

The gas emission factor is 56.1 kg CO_2 GJ^{-1}. Methane combustion uses 44.01/16.04 kg CO_2 per kilogram of methane. Upstream ethane is assigned 0.50 kg CO_2e kg^{-1} ethane. The avoided-steam factor is 56.1/0.90 kg CO_2 GJ^{-1}, and the avoided-C_3+ factor is 0.80 kg CO_2e kg^{-1} C_3+. All plotted cases use these factors. With eventual combustion of all fuel gas and system-expansion credits, the carbon intensity in kg CO_2e per tonne ethylene is

$$\begin{aligned} \text{CI} = {} & 10^{-3} q_{\text{buy}}(56.1) + 1000 m_{\text{CH}_4} \frac{44.01}{16.04} + 1000 m_C \frac{44.01}{12.011} + E_t e_g \\ & + 1000 m_{\text{fresh}} e_A - 10^{-3} q_{\text{TLE}} \frac{56.1}{0.90} - 1000(m_{\text{C}_3} + m_{\text{C}_{4+}})(0.80) \end{aligned} \tag{S26}$$

Here e_g is the grid emission factor in kg CO_2e kWh^{-1} and $e_A = 0.50$ kg CO_2e kg^{-1} ethane. The coke term is zero for GP chemistry. Methane burned in the furnace and methane exported as fuel gas sum to the full methane production within this system boundary. At the same coil effluent composition and heat demand, production cost and CI vary linearly with electricity price and grid emission factor, respectively. Writing $\Delta C_E = \Delta C_0 + \Delta E_t p_e$ and $\Delta\text{CI} = \Delta\text{CI}_0 + \Delta E_t e_g$, with electric minus fired differences, gives the break-even electricity price $p_e^* = -\Delta C_0/\Delta E_t$ and the break-even grid emission factor $e_g^* = -\Delta\text{CI}_0/\Delta E_t$ when $\Delta E_t \neq 0$. The zero-subscript terms contain all terms independent of electricity price or grid emission factor.

The carbon accounting omits construction, upstream purchased-gas emissions, and a separate burden for dilution-steam generation.

S4. Reproduction

The implementation corresponding to Sections S1–S3 is organized as follows: `multiscale/cantera_sweep.py`, function `simulate_case`, defines the segmented reactor and output basis. `multiscale/train_gp.py` defines transformations, tuning, fitting, and uncertainty calibration. `multiscale/gp_runtime.js` evaluates full surrogate predictions and uncertainties. The `mechanismGP`, `model`, and `lca` functions in the repository-root `engine.html` implement the process point predictions, utility and economic calculations, and carbon accounting. The serialized `multiscale/surrogate.json` contains the normalization constants, training coordinates, coefficient matrix, inverse covariance, and output calibration factors needed to reproduce released predictions without retraining.

The reference case is evaluated with both GP and empirical chemistry. The gas temperature–time profile, electricity-price, and grid-emission-factor sweeps use GP chemistry. Test-set predictions are compared with the Cantera calculations, and element and mass closure are checked for the Cantera calculations. Direct Cantera calculations are repeated at R, A, and B, including a 40-segment refinement at B. The training set contains 640 conditions, and the segment-convergence comparison covers five conditions.

S5. Comparison with published process cases

The literature comparisons use the surrogate described in Section S1 and the published quantities transcribed in `benchmarks/literature.json`. The effluent composition of Wang et al. used in Figure 4d is taken from Table 5 of the source article.

For the Chen et al. case (Green Chemistry, 2024), residence time is fitted to a residual ethane dry mole fraction of 0.151 at 950 °C. The two reported pressure interpretations, 1 and 4.5 bar, give residence times of 0.101–0.132 s. Ethylene and hydrogen dry mole fractions are 0.372–0.385 and 0.399–0.412, against reported values of 0.377 and 0.407. Fresh ethane demand is 1.185–1.219 versus 1.108 kg kg^{-1} ethylene. The calculated enthalpy rise from 650 °C to the COT is 7.60–7.68 GJ t^{-1}, and the reported reactor heat is 5.25 GJ t^{-1}. Interpreting this difference requires the same temperature limits for both values.

For the Shin et al. case (Green Chemistry, 2025), residence time is fitted to fresh ethane demand over 800–900 °C and 1–5 bar. Forty-three operating conditions match the fresh-feed demand within 0.5%, with residence times of 0.666–1.50 s. At the default process parameters, the calculated combustion heat (LHV) and direct (scope 1) emissions are 17.20–18.58 GJ t^{-1} and 415–527 kg CO_2e t^{-1}, compared with 17.2 and 442 reported. C_3+ production is 0.1097–0.1273 versus 0.1376 kg kg^{-1} ethylene. The subsequent uncertainty analysis uses Shin's combustion heat and Chen's compressor work to calibrate effective utility parameters.

Wang et al. (Frontiers in Energy Research, 2026) report the tube geometry and feed flow rate, which correspond to a residence time at mean gas density of 0.291 s. At 840 °C and 3.2 bar, the maximum single-pass conversion in the tested heating-exponent range is 0.366 at $n = 0.45$, against 0.685 reported. The paper uses a different kinetic model, a 502.77 °C inlet, and a feed containing 0.64 wt% ethylene, whereas the surrogate calculations use a 650 °C inlet and an ethane–steam feed. Economic outputs are reported for context only because capital, financing, and coproduct boundaries differ among studies. For Figure 4d, the mass fractions of the reactor effluent stream (P-OUT) in Table 5 of Wang et al. define the published effluent composition, including an ethylene mass fraction of 0.3995. The acetylene hydrogenation reactor model is applied to both effluent compositions, and the results are normalized by their final ethylene production. The dry effluent includes ethane, ethylene, hydrogen, methane, acetylene, C_3, and C_{4+}, and steam is removed before

compression. C_3 combines propylene and propane, and C_{4+} combines the listed C_4 species and benzene. Both compositions are evaluated with the equations in Section S2, including compression from 1.5 to 32 bar, the same acetylene-hydrogenation assumptions, and common recovery and purity specifications. Compression, cold-box, and C_2-splitter electricity costs are 20.0, 31.1, and 24.8 USD t^{-1} for the published composition and 27.7, 44.9, and 43.1 USD t^{-1} for the GP-predicted composition. These costs quantify the effect of effluent composition with the same separation model. The comparisons of five effluent species are given in the accompanying comparison data files.

For the conventional case of Mittal et al. (Chemical Engineering Journal, 2025), the source SI gives a reactor temperature of 860 °C and a pressure of 1 bar. The source states dilution as an ethane-to-steam volume ratio of 3:1 (0.20 kg steam per kg ethane) in its SI and as 30 wt% steam (0.43 kg steam per kg ethane) in its main text, and both values are evaluated. The calculation uses $n = 1$, fired heating at 60% efficiency, 616 kt y^{-1} ethylene, 200 USD t^{-1} ethane, and 550 USD t^{-1} LPG. Residence time fitted to the reported fresh ethane demand of 1.19 kg kg^{-1} ethylene is 0.77–1.37 s, with single-pass conversion of 0.633–0.769. The source life-cycle inventory lists 3.41 kWh kg^{-1} electricity, whereas its electricity expense of 37.6 million USD y^{-1} corresponds to 0.87–1.02 kWh kg^{-1} ethylene at 6–7 cents/kWh. Industrial electricity averaged 6.12 and 5.61 cents/kWh in Texas and Louisiana in 2024 (EIA Electric Power Annual, Table 2.10), and the comparison uses the expense-based value. The capital correlation is scaled from this case's reported investment of 317 million USD, so the calculated TCI of 347–386 million USD is a check of the capital scaling. The calculated production cost (383–415 USD t^{-1}) and carbon intensity (1.16–1.26 kg CO_2e kg^{-1} ethylene) are omitted from the comparison with the reported 680 USD t^{-1} minimum selling price and 4.12 kg CO_2e kg^{-1} ethylene because utility prices, coproduct credits, and LCA allocation differ.

S6. Surrogate validation

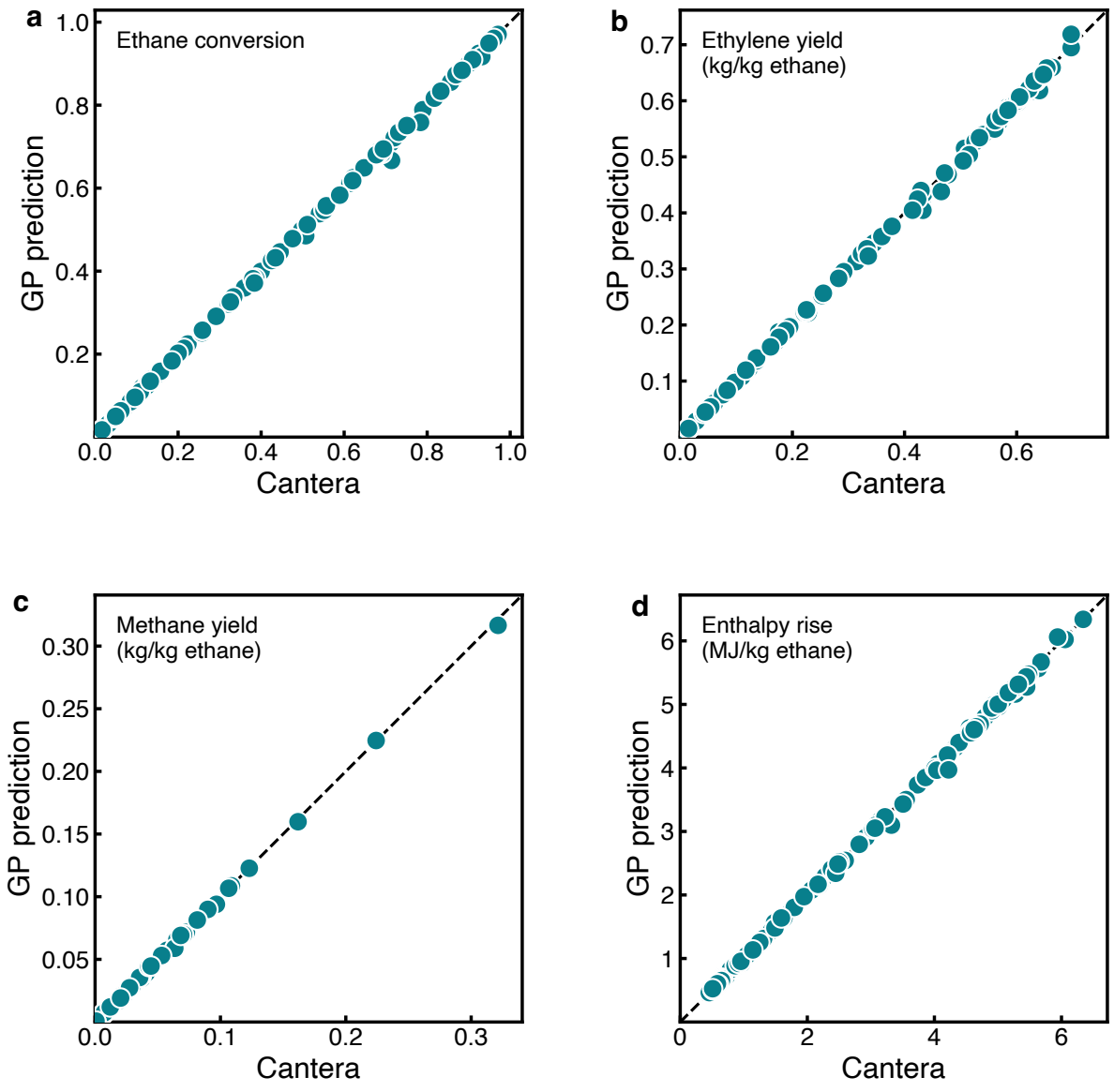


Figure S1. Surrogate predictions against independent Cantera calculations. **(a)** Single-pass ethane conversion, **(b)** single-pass ethylene yield, **(c)** methane mass yield, and **(d)** enthalpy rise for 128 Cantera calculations excluded from training. Symbols are GP predictions and lines show parity. Mass yields and enthalpy rise are per kg coil-inlet ethane.

S7. Browser application

Figure S2 shows the released browser application for the reference fired case. One screen shows the reactor inputs, the process flowsheet, and the calculated production cost and carbon intensity.

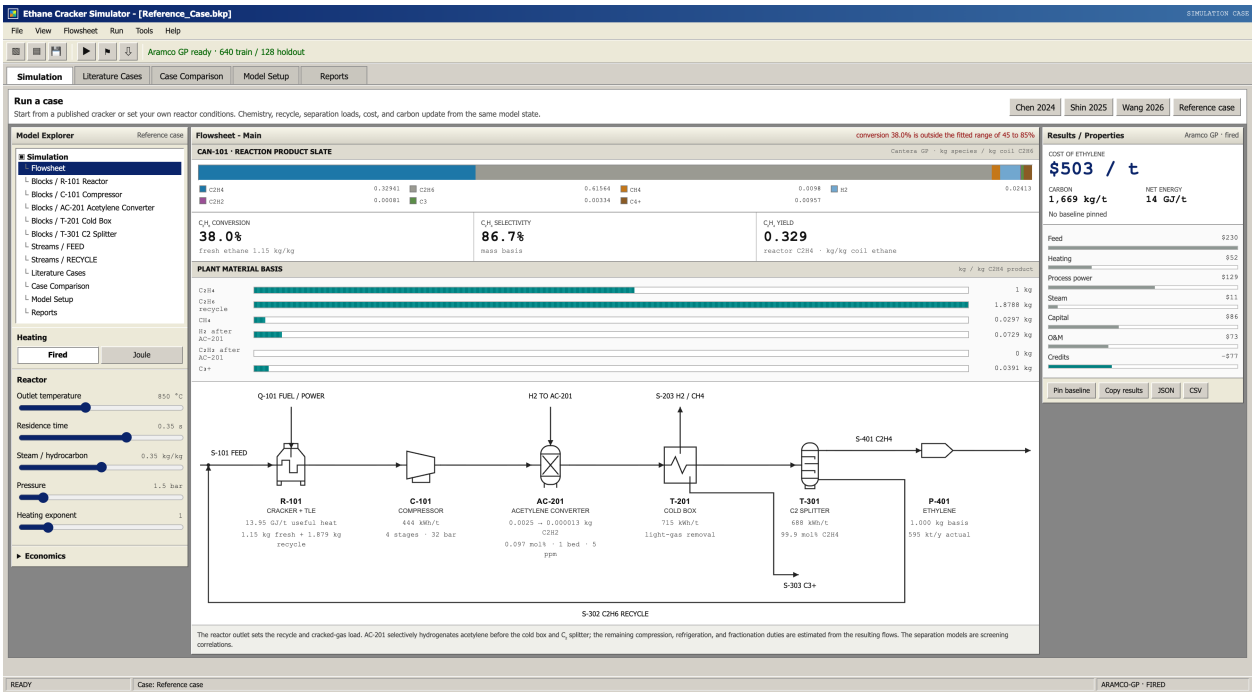


Figure S2. Browser application for the reference fired case. Inputs are a COT of 850 °C, 0.35 s, a dilution steam ratio of 0.35 kg steam per kg ethane, 1.5 bar, and heating exponent 1. The application includes the acetylene hydrogenation reactor and reports a production cost of 503 USD t^{-1} ethylene, the same value as the reference case in Table 2.